ICU Mortality Risk Prediction

Using Routine ICU Data to Estimate Hospital Mortality

Kaggle MIMIC-Inspired ICU Dataset · Logistic Regression & XGBoost · n = 15,000

AI in Medicine Course Project

Project Overview

01

Objective

Build a mortality risk prediction tool using routinely collected ICU data to help clinical teams identify patients who may need closer monitoring or earlier escalation.

02

Dataset

15,000 ICU patient records from Kaggle. Features include age, comorbidity scores, vital signs, oxygenation, sepsis status, ventilation, and vasopressor use. Mortality rate: 22.8%.

03

Approach

Two models compared — Logistic Regression (interpretable) and XGBoost (flexible). Decision thresholds were optimised for clinical utility (F1-score). The tool is a clinical support aid, not a decision-maker.

Dataset & Class Distribution

Key Dataset Stats

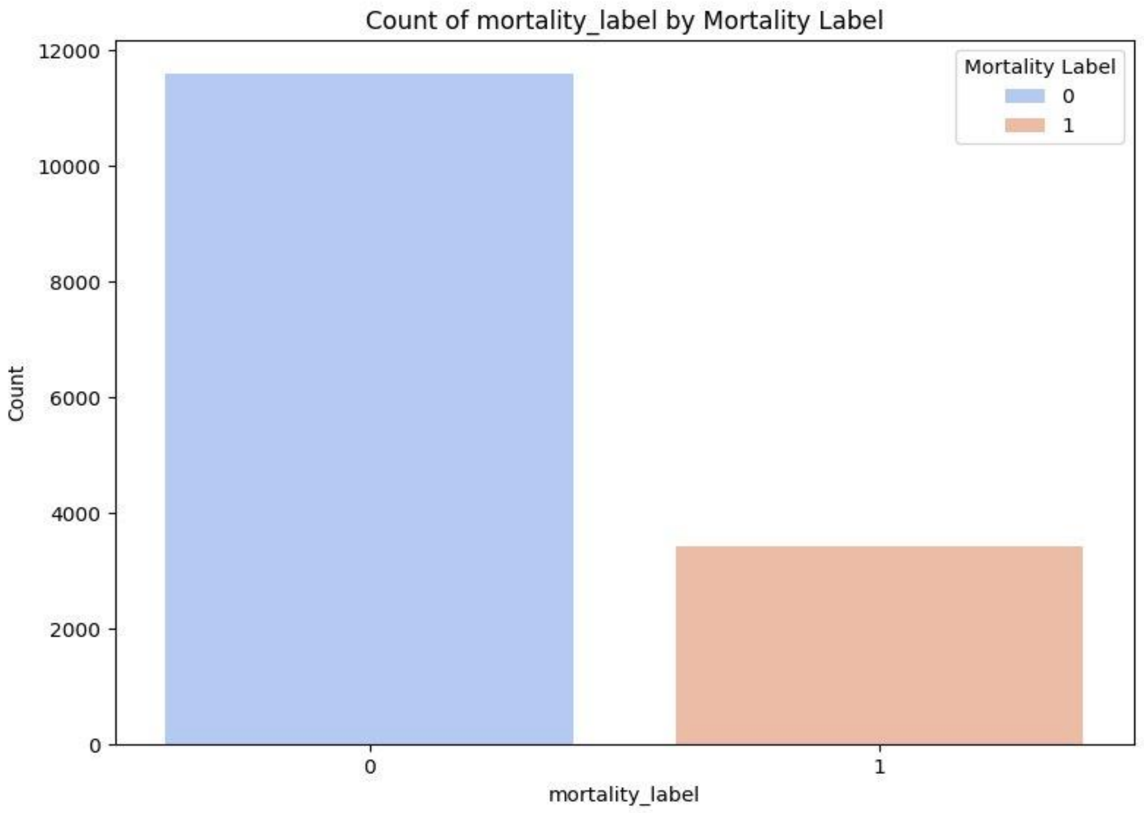
15,000
Total Patients

22.8%
Hospital Mortality

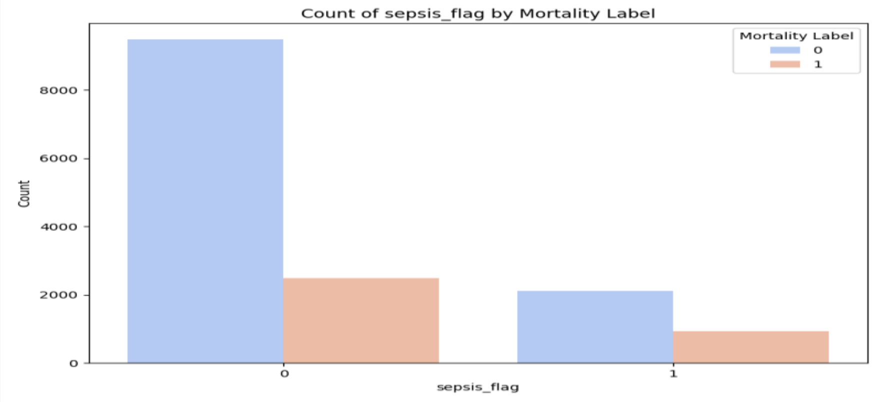
~11,600
Survivors (Label 0)

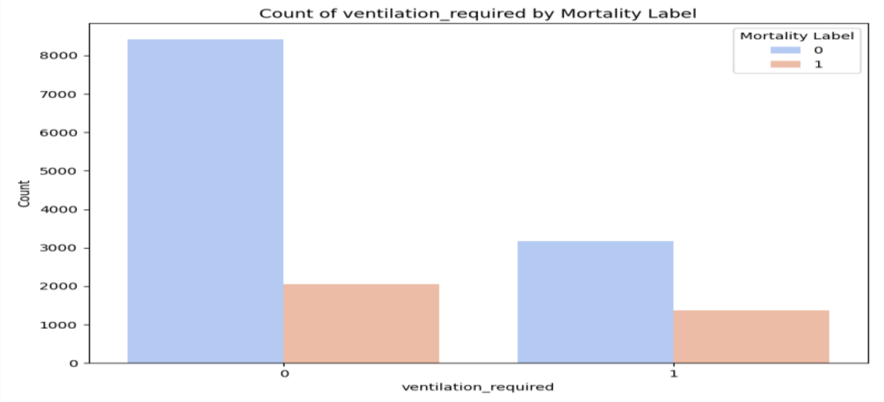
~3,400
Non-survivors (Label 1)

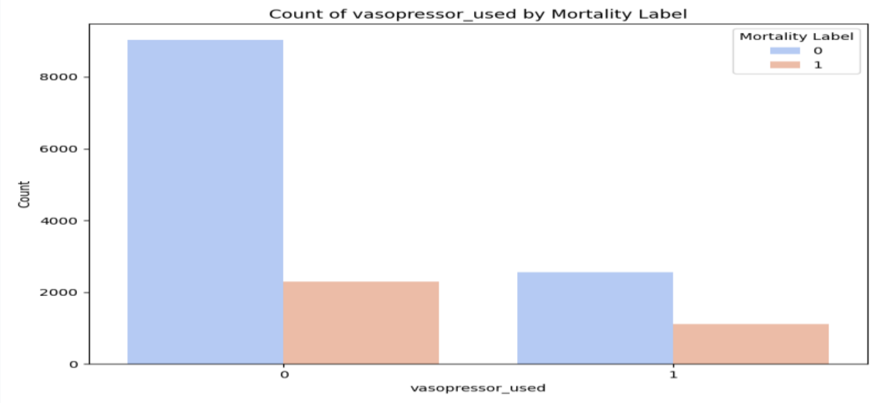
20+
Features

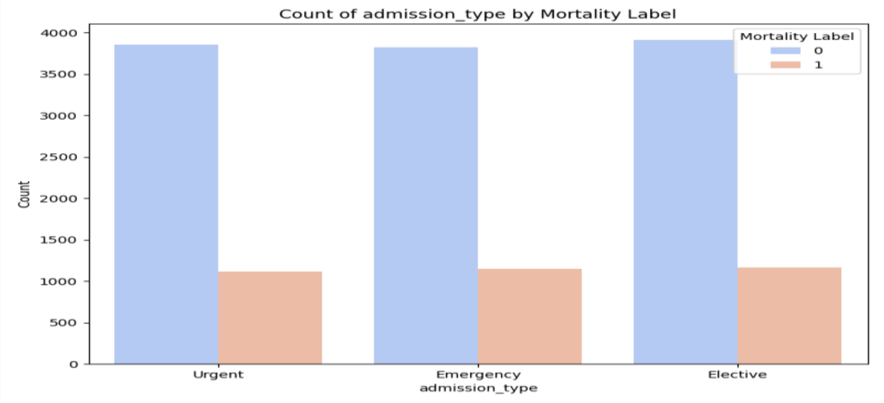


Exploratory Data Analysis: Clinical Risk Factors



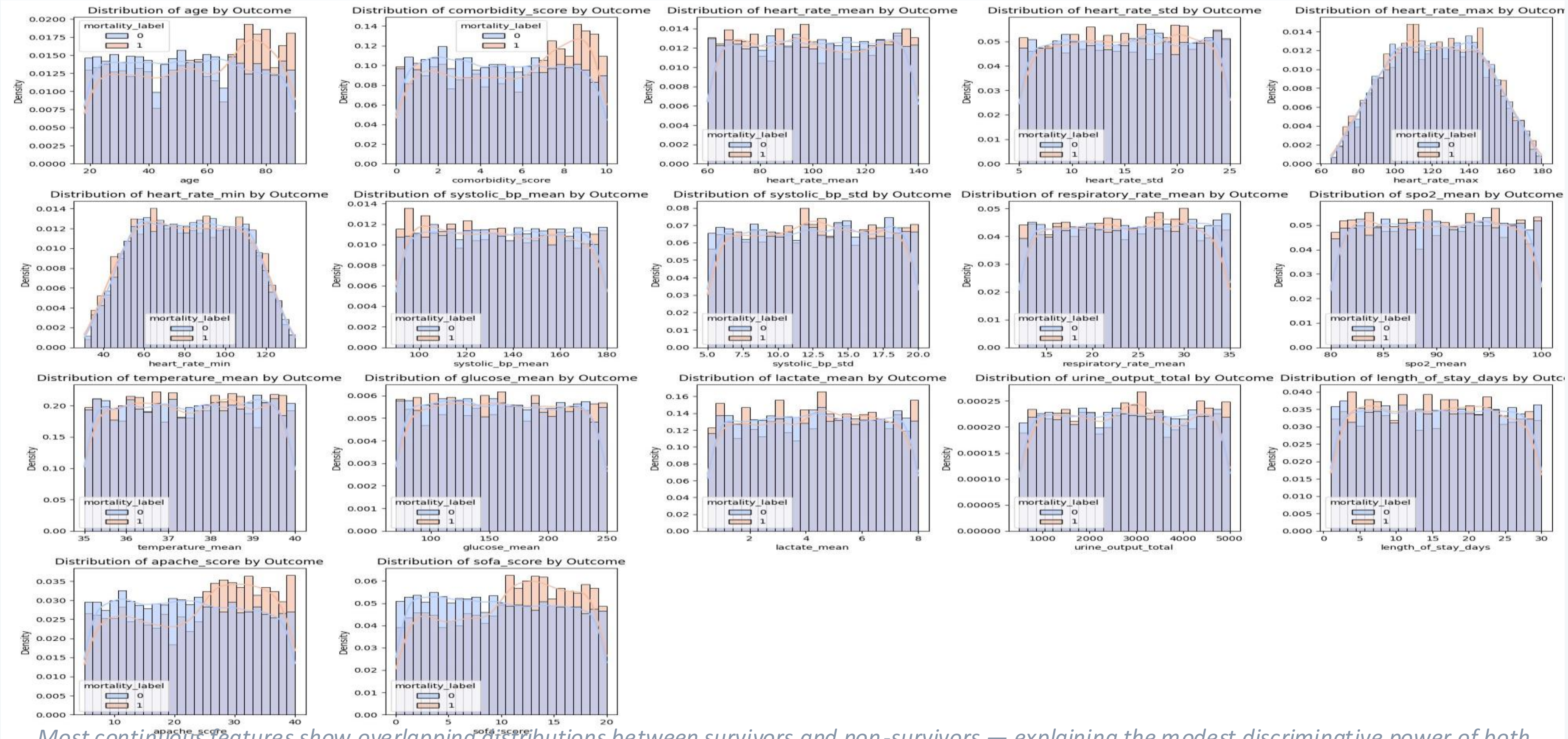






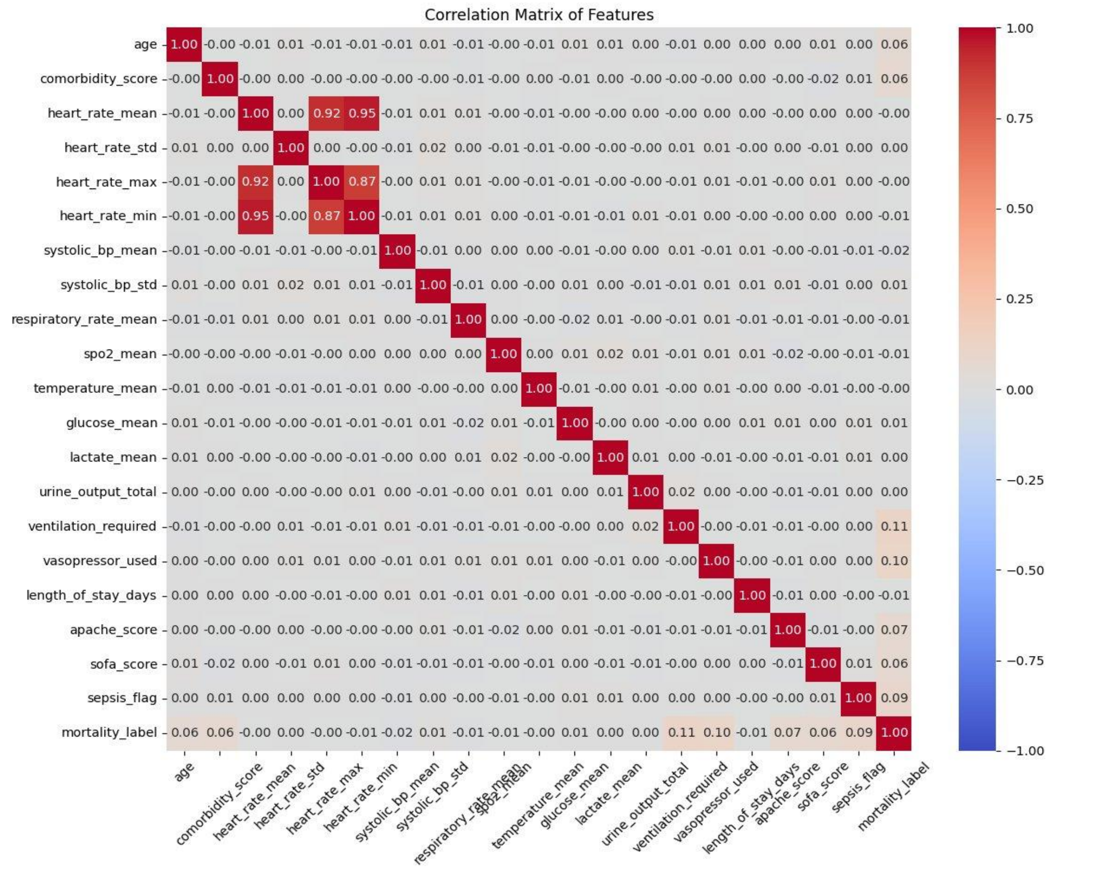
Sepsis, ventilation, and vasopressor use show the strongest associations with ICU mortality

Feature Distributions by Outcome



Most continuous features show overlapping distributions between survivors and non-survivors — explaining the modest discriminative power of both models.

Feature Correlation Matrix



Weak signal overall

Most features show near-zero correlation with mortality_label, confirming limited predictive power in the dataset.

Heart rate cluster

heart_rate_mean, max, and min are highly inter-correlated ($r \geq 0.87$) — expected from the same vital sign.

Strongest predictors

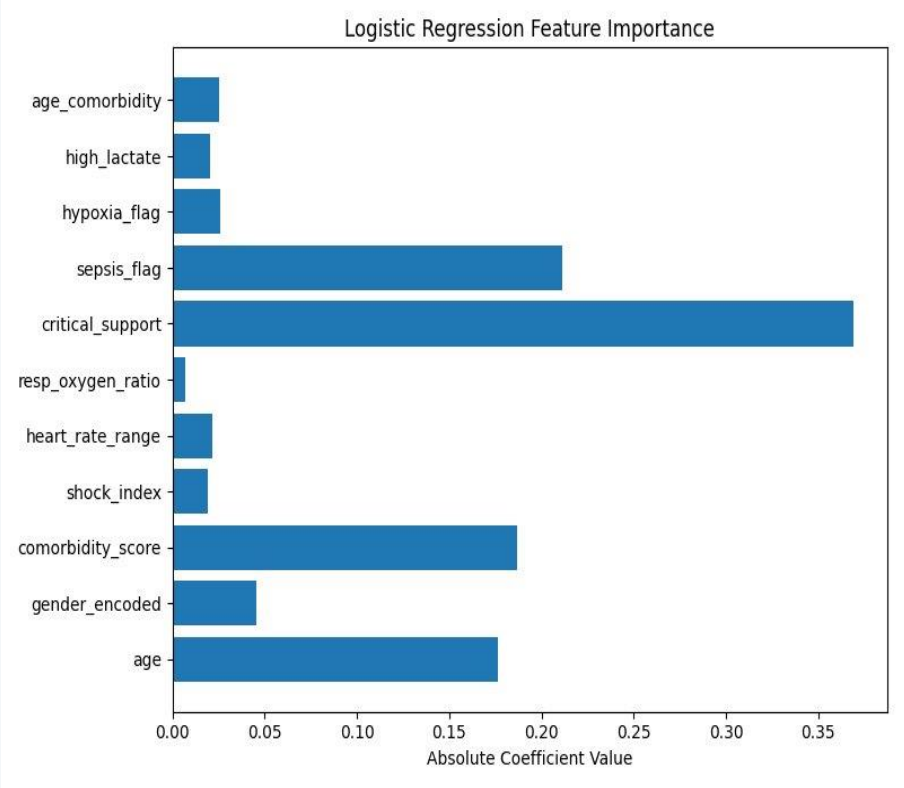
ventilation_required (0.11) and vasopressor_used (0.10) have the highest correlation with mortality, consistent with EDA findings.

Sepsis association

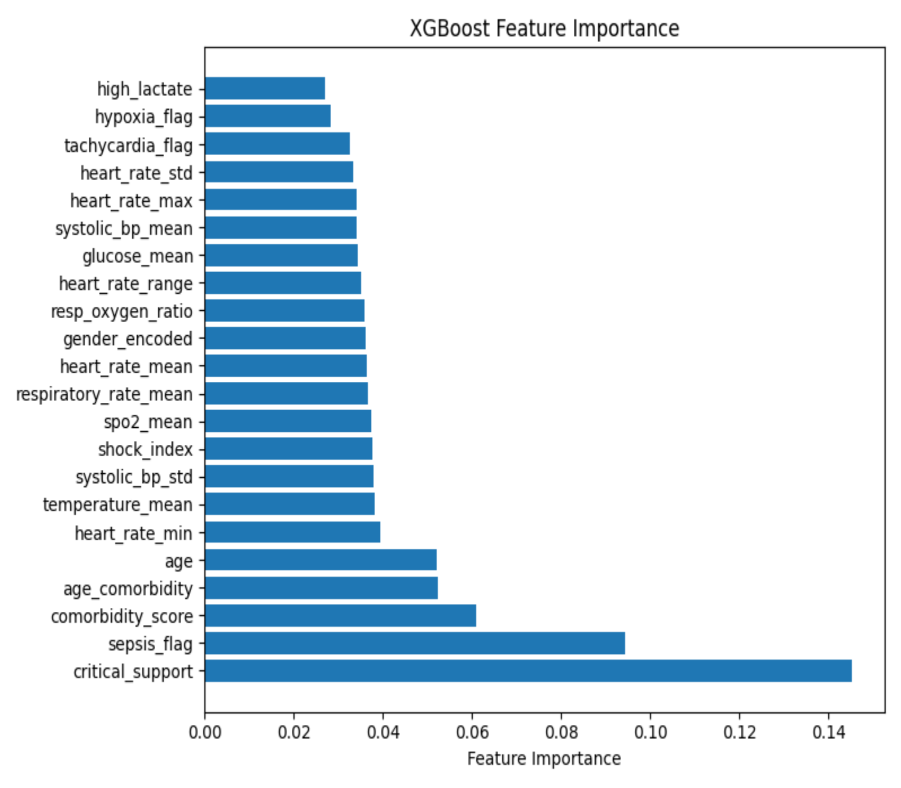
sepsis_flag correlates modestly with mortality (0.09) and with vasopressor use — as expected clinically.

Feature Importance: What Drives Risk?

Logistic Regression

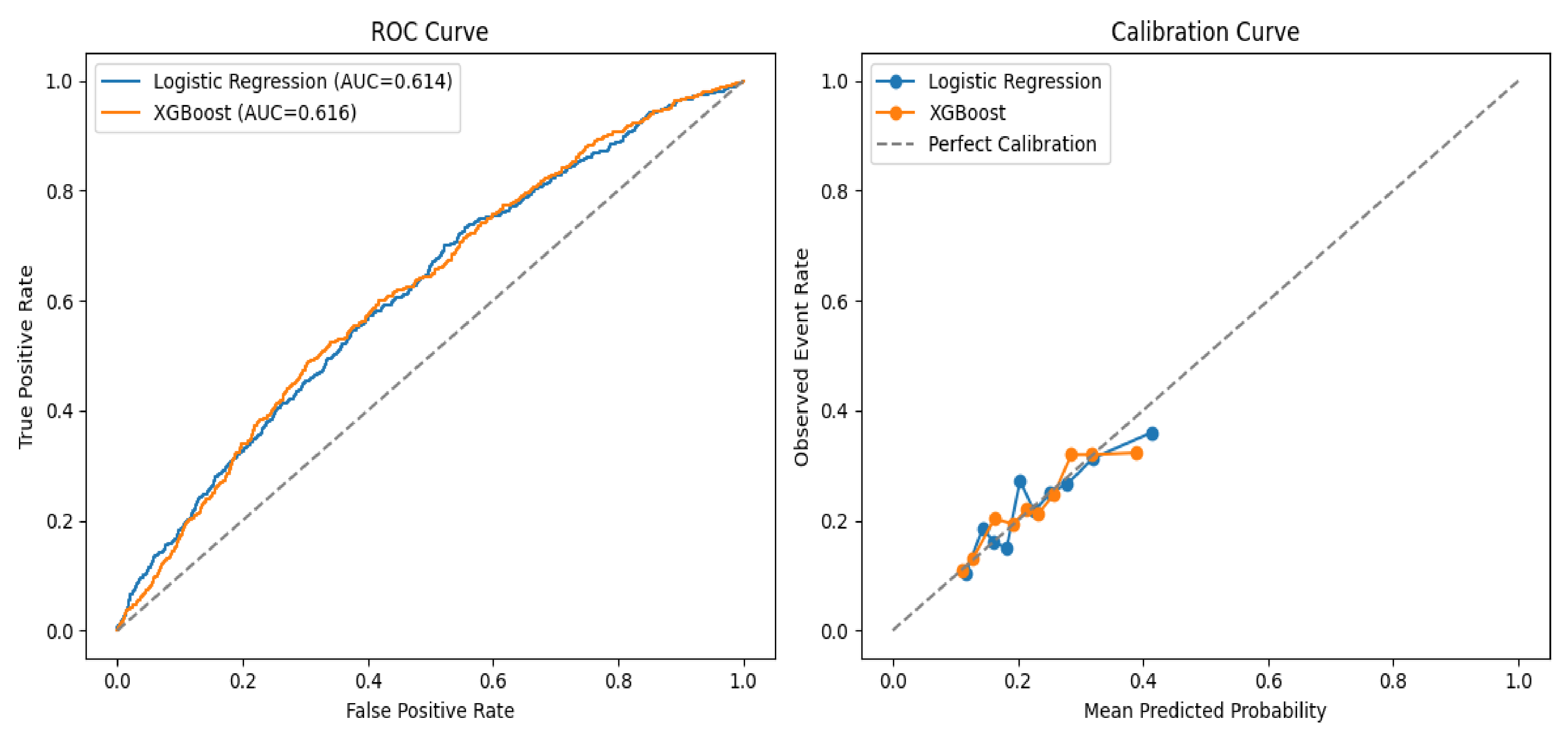


XGBoost



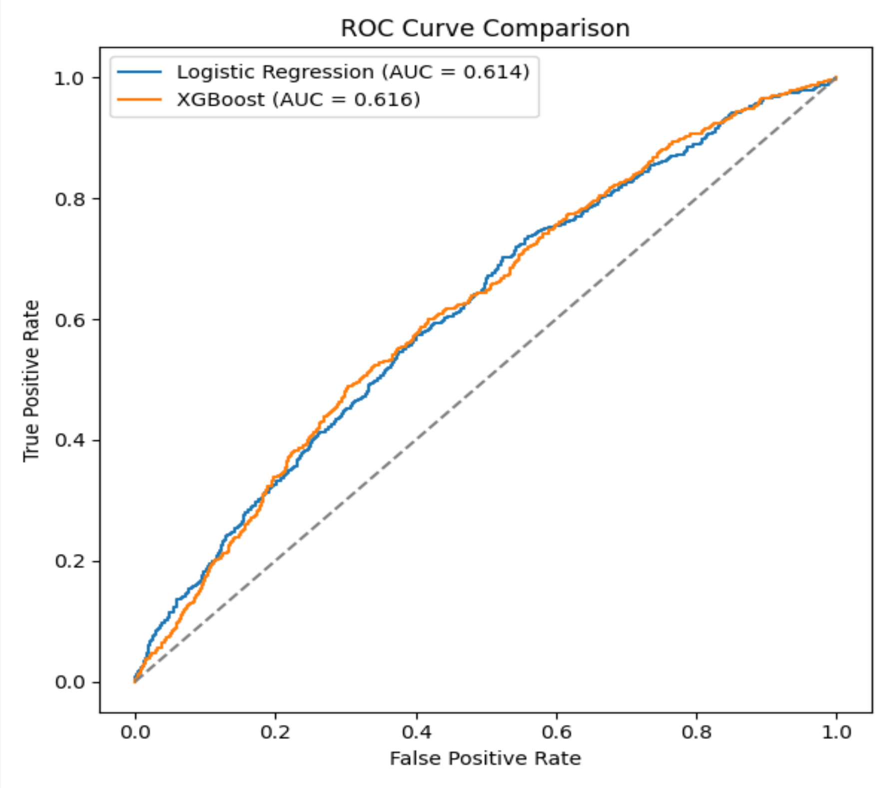
Both models agree: critical_support and sepsis_flag are the dominant predictors of ICU mortality.

Model Performance Overview

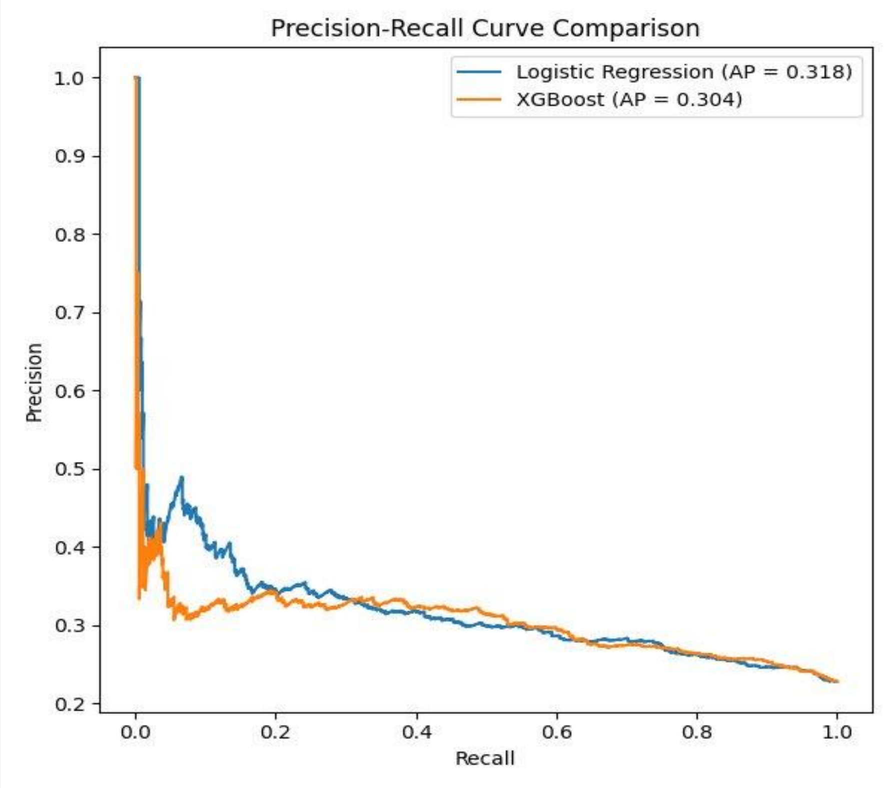


Discriminative Performance: ROC & Precision-Recall Curves

ROC Curve (AUC ~0.61)

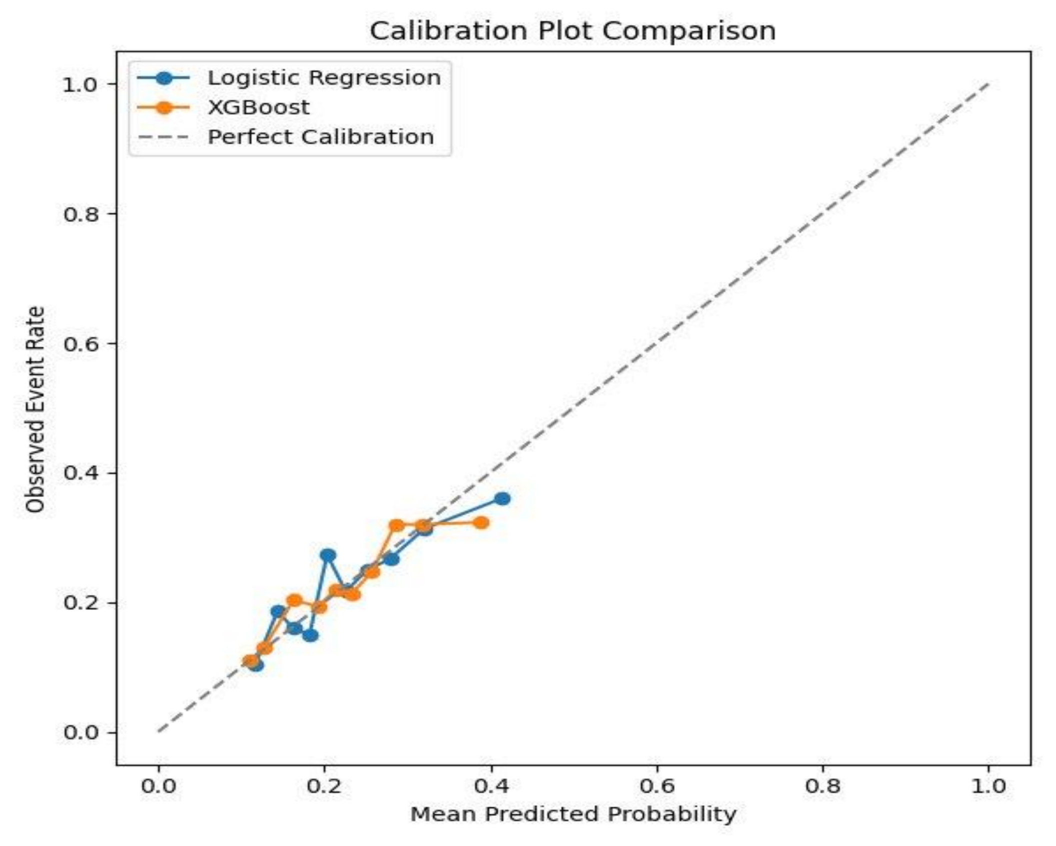


Precision-Recall Curve (AP ~0.31)



Both models perform modestly above chance. The low PR-AUC reflects the challenge of the 22.8% minority class.

Probability Calibration



Brier Score

0.171

(Both models — lower is better)

What this means

A Brier score of 0.171 indicates excellent probability calibration. When the model predicts 35% mortality risk, approximately 35% of those patients truly die.

Why this matters clinically

Even if a model cannot perfectly rank patients, well-calibrated probabilities are directly usable — clinicians can trust the raw numbers as genuine risk estimates, not just relative rankings.

The key insight

Both models are mathematically trustworthy in their predicted probabilities, even when discriminative power is modest.

The Clinical Trade-Off: Sensitivity vs. Alarm Fatigue

Metric	Logistic Regression	XGBoost
AUROC	0.614	0.616
PR-AUC	0.318	0.304
Precision	0.280	0.270
Recall	**0.737**	**0.773**
Brier Score	0.171	0.171

Wide Safety Net (High Recall ~75%)
Models catch ~75% of patients who die. XGBoost is slightly more aggressive at 77%.

False Alarm Cost (Low Precision ~27%)
Only ~27–28% of flagged patients actually die. In a busy ICU, this risks alarm fatigue.

Logistic Regression Preferred

Equivalent performance — but far easier to interpret and explain at the bedside.

Interpreting the Risk Score

0–30% Lower Risk	Do not assume the patient is safe. Compare with the bedside picture and recent trend.
30–70% Intermediate Risk	Use caution. Clinical judgment is especially important at this threshold.
70–100% Higher Risk	Correlate with the clinical picture. Consider closer review and possible escalation if supported clinically.

Before Acting on Score

? Does the result match the patient's current condition?

? Are the input data complete and accurate?

? Is the patient worsening despite a low score?

? Is the patient stable despite a high score?

The score supports — it does not replace — clinical judgment.

Limitations & Failure Modes

Single Dataset Only

Tested on one internal ICU dataset from Kaggle. Performance in other hospitals, health systems, or patient populations is unknown.

Modest Discriminative Power

AUROC ~0.61 is only slightly better than chance. Both models should be understood as risk-awareness tools, not diagnostic instruments.

False Positives & Alarm Fatigue

Precision ~27% means roughly 73% of high-risk flags are incorrect. In a high-acuity setting, this creates a meaningful alarm fatigue risk.

False Negatives & False Reassurance

Recall ~75% means ~25% of patients who die are not flagged. A low score must never be used to deprioritize a patient showing clinical decline.

No Subgroup Validation

Subgroup analyses (age, diagnosis, admission type) and comparison with clinician performance were not performed.

Data Quality Dependency

Performance degrades with missing, delayed, inaccurate, or non-representative input data.

Human Override & Clinical Workflow

SITUATION	ACTION
High score, low clinical concern	Recheck data and reassess patient. Do not escalate on score alone.
Low score, high clinical concern	Follow the bedside picture. Do not let a low score override concern about deterioration.
Missing or unreliable data	Do not rely on the score until the data are verified.
Patient outside validated population	Use extra caution. Treat result as supplemental information only.
High-stakes decisions	Never use the score alone for major treatment or goals-of-care decisions.

The tool fits best into routine ICU care as an added risk-awareness step during admission, rounds, handoff, or changes in patient status.

Conclusions

01 Both Logistic Regression and XGBoost show modest but comparable discriminative performance (AUROC ~0.61), constrained by weak predictive signals in the dataset.

02 Threshold optimisation produced sensitive safety-net models (~75% recall) — but with meaningful false-positive rates (~73%) that risk alarm fatigue in the ICU.

03 Both models are well-calibrated (Brier ~0.171), meaning their predicted probabilities are genuinely trustworthy as risk estimates.

04 Logistic Regression is the preferred model: equivalent performance, superior interpretability, and easier to explain to clinical teams.

05 External validation, subgroup analysis, and prospective clinical testing are required before any real-world deployment.